

Code-Layout, Readability and Reusability

Date	15 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4-space indentation maintained throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Source code, templates, static files, models, and dataset are organized into separate folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables like temperature, humidity, rainfall, prediction, and model are descriptive.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as home(), about(), and predict() clearly indicate their purpose.
5	Code Comments	Inline and block comments explain logic	Partial	Important sections are commented; additional comments can improve readability.
6	Modular Design	Code is split into reusable functions/modules	Yes	Application, model, templates, and static resources are separated into modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Repeated code is minimized, ensuring maintainability.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling are implemented in the prediction module

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Flask Application (app.py)	Python, Flask	Handles routing and web requests	High
2	Machine Learning Model (model.pkl)	Scikit-learn	Used for crop prediction	High
3	Prediction Module	Python	Used to process user inputs and generate predictions	High
4	HTML Templates	HTML, Bootstrap	Shared across Home, About, and Prediction pages	Medium
5	CSS Stylesheet (style.css)	CSS	Used throughout the web application	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized folder hierarchy with separate modules for source code, templates, models, and datasets.
Readability	5	Descriptive variable names, functions, and consistent formatting improve readability.
Reusability	5	Reusable modules, templates, CSS, and trained ML model reduce duplication.
Documentation / Comments	4	Core sections are documented; additional inline comments can further improve maintainability.
Overall Score	4.8 / 5	Clean, modular, maintainable, and suitable for future enhancements.